

Amresh Yadav

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EDUCATION

Indian Institute of Information Technology, Ranchi

Bachelor of Technology in Electronics and Communication Engineering | CGPA: 9.64/10

Ranchi, Jharkhand

11/2022 – 05/2026

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C/C++, SQL, HTML/CSS

Frameworks & Libraries: React.js, Next.js, Node.js, Express.js, FastAPI, Flask, Redux Toolkit, Tailwind CSS, Material UI

Databases & Messaging: PostgreSQL, MongoDB, MySQL, Redis, Snowflake, Apache Kafka, gRPC

Cloud & DevOps: AWS (EC2, S3, Amplify), Docker, Microservices, GraphQL

AI Tooling: Cursor, Claude Code, GitHub Copilot, LLM/GenAI API integration (GPT, Gemini)

Core CS: Data Structures & Algorithms, System Design, Object-Oriented Programming (OOP), DBMS, Operating Systems, Computer Networks, Distributed Systems

EXPERIENCE

Ocrolus

12/2025 – 08/2026

Software Engineer Intern | Python, FastAPI, Flask, gRPC, React.js, TypeScript, Kafka, Redis, Snowflake, PostgreSQL Gurugram, India

- Migrated internal ops tools from **Retool** to a custom **React.js + TypeScript** platform backed by a **Flask/FastAPI** BFF and **gRPC** microservices, saving **\$100K+** in annual licensing and infrastructure cost.
- Eliminated direct SQL access from internal tools by routing all traffic through an auth-enforcing BFF and typed gRPC contracts, hardening security, data isolation, and service decoupling across distributed systems.
- Developed an **LLM-driven transaction tagging** system (GPT-4o-mini for recall, Gemini for verification) over **Snowflake** with an automated pipeline that discovers candidate tags and impact-measures each before rollout.
- Layered a two-gate, **four-eyes (maker-checker)** human review workflow on top of the pipeline, safely expanding tag coverage without corrupting customer financial analytics.
- Architected an **async re-enrichment pipeline** on **Apache Kafka** with book-keyed partitioning for ordering, an isolated crisis-lane topic, and **DLQ**-based failure recovery, decoupling heavy background work from a blocking-free UI.
- Reduced read-endpoint latency by introducing distributed **Redis** caching with event-driven invalidation, **PostgreSQL** query tuning, and leaner **gRPC** payloads across analytics services.
- Revamped the client-facing **Account Summary** dashboard in **client.ui (React.js)** from **Figma** designs, shipping a pixel-accurate, responsive UI now in production use by clients.

Saarthee

06/2025 – 07/2025

Full Stack Developer Intern | React.js, FastAPI, Node.js, Redux Toolkit, Docker, AWS

Gurugram, India

- Crafted **10+ reusable React.js components** for an algorithmic trading platform with a modular, prop-driven architecture, improving page load speed by **30%** and accelerating feature development across the team.
- Integrated **15+ RESTful APIs** with **Redux Toolkit** state management, reducing bug reports by **25%**, and designed a dynamic column-matching tool aligning **500+ financial files** at **95% accuracy**.
- Containerized the full stack with **Docker** for dev/staging/prod parity and deployed via **AWS Amplify, EC2, and S3**, supporting **1,000+ daily file operations** with secure file handling.

DraconX

03/2025 – 05/2025

Full Stack Developer Intern | Python, Node.js, Express.js, React.js, PostgreSQL, GenAI APIs

Hyderabad, India

- Contributed to an **AI-driven RFP automation platform**, building Proposal Tracking, Compliance Check, and RFP Discovery modules that accelerated proposal drafting and streamlined compliance review for enterprise clients.
- Embedded **Generative AI** APIs into the **Node.js/Express** backend with structured prompt workflows, enabling automated drafting, smart content generation, and real-time compliance feedback in React collaborative views.

FandomX

10/2024 – 01/2025

Full Stack Developer Intern | React.js, Node.js, Express.js, MongoDB, JWT

Bengaluru, India

- Launched FandomX's first web platform (a sports-centric ticketing and e-commerce app) with a responsive **React.js** frontend and secure **JWT/bcrypt** authentication with Zod validation.
- Implemented core MVP features including ticket booking, venue reservation, and e-commerce flows with **EmailJS**-powered email communication, establishing the platform's initial online presence.

PROJECTS

Wanderlux — Hotel Booking System | Node.js, gRPC, Kafka, PostgreSQL, Redis, Next.js, Docker



- Built **4 microservices + an API Gateway** (Auth, Hotel, Booking, Payment) with schema-per-service **Postgres**, **JWT** auth, and **Redis** distributed locks and caching for safe concurrent room inventory.
- Shipped a **Kafka** choreography saga with a transactional outbox for reliable booking and payment confirmation, plus compensating cancel and refund flows to keep services consistent.

AmiChat — Realtime Chat App | React.js, Node.js, Express.js, MongoDB, Socket.io



- Created a real-time chat application using **Socket.io** with **100% multi-device** functionality and secure **JWT/bcrypt** authentication, integrating **Cloudinary** for image uploads and optimizing the backend with **MongoDB** and **Express.js**.

ACHIEVEMENTS

- Attained **LeetCode Knight** badge with a peak contest rating of **1872**, solving **800+** problems, and **400+** on **GeeksforGeeks**.
- Achieved **3-Star** on **CodeChef** (peak **1625**, Global Rank 343 in Starters 166) and a peak rating of **1363** on **Codeforces**.
- Secured **All India Rank 1 & 7** in EROH (organized by Naukri.com) among **20,000+ participants** from IITs, NITs, IIITs, BITS, and other top engineering institutes.